"A properly doped crystal diode which has a sharp breakdown voltage is known as a zener diode."

It may be seen that it is just like an ordinary diode except that the bar is turned into z-shape.

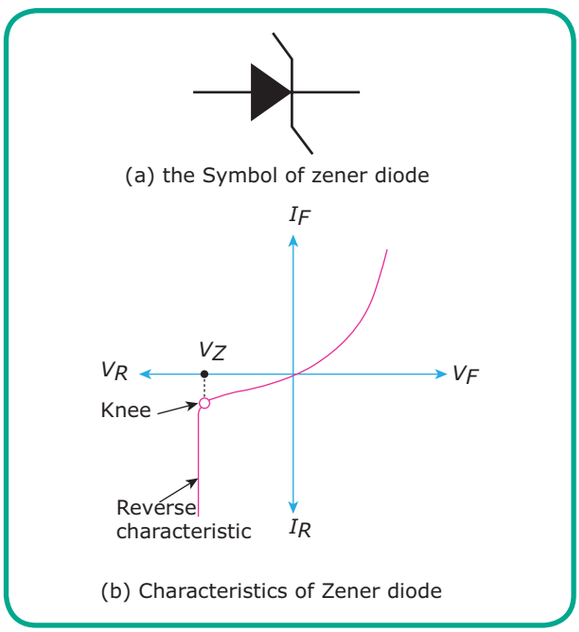


Figure 4.21 Symbol and characteristics of zener diode

The figure 4.21(a) shows the symbol of Zener diode. Figure 4.21(b) shows the characteristics of Zener diode. The breakdown or zener voltage depends upon the amount of doping. If the diode is heavily doped, depletion layer will be thin and consequently the breakdown of the junction will occur at a lower reverse voltage. On the other hand, a lightly doped diode has a higher breakdown voltage.



Who discovered zener diode? Clarence Melvin Zener.

Clarence Melvin Zener (December 1, 1905 – July 2, 1993) was the American physicist who first (1934) described the property concerning the breakdown of electrical insulators. These findings were later exploited by Bell Labs in the development of the Zener diode, which was duly named after him.



Clarence Melvin Zener.

The following points may be noted about the zener diode:

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1. A zener diode is like an ordinary diode except that it is properly doped so as to have a sharp breakdown voltage
2. A zener diode is always reverse connected i.e. it is always reverse biased.
3. A zener diode has sharp breakdown voltage, called zener voltage V_Z.
4. When forward biased, its characteristics are just those of ordinary diode.
5. The zener diode is not immediately burnt just because it has entered the breakdown region. As long as the external circuit connected to the diode current to less than burn out value, the diode will not burn out.

4.9. VOLTAGE DOUBLER

Voltage doubler is the circuit where we get twice the input voltage, like if we supply 10 V, we will get 20 V at the output. Generally, transformers are there to step-up or step-down the voltage but sometimes transformers are not feasible of their size and cost. The quick, easy and practical solution is to double the voltage using a voltage doubler. A full-wave voltage doubler is a voltage multiplier with a multiplication factor of two. A full-wave voltage doubler is shown in the Figure 4.22.

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